ALIoT Smart Lab - Continuation Report

17 October 2026 (evening) · Portal **http://192.168.1.200** (browser screen sharing needs **https://192.168.1.200**) · nothing committed or pushed · `smartlab_restore_drill` kept

1. Results by task

Task	Done	Measured / tested	Status
PC screen capture (missing Pillow)	The agent now captures without Pillow, using Windows' own screen copy (GDI) plus OpenCV, which is already installed for the cameras. Pillow is used when present. The agent is DPI-aware, so monitors report real pixels (1920×1080, not 1536×864). Nothing was installed.	5 agent tests. Real frames from both monitors: 80-150 ms per frame.	**VERIFIED**
PC preview, monitors, wall, reconnect, sharing	**Live test:** the real agent ran on this laptop as the admin PC (`PC_ADMIN_LAPTOP` , master PC). **Agent poll:** 2 s → 1 s. **Bug fixed:** "View all screens" was hidden when the only PC was the master PC. **Pilot-install files:** `start_pc_agent.bat` , `agent.env.example` ; `agent.env` is git-ignored.	Preview: first frame 1.7 s (was 2.5 s), 2.2 frames/s. Monitor 1 / 2 / all = 1280×720 / 1280×800 / 1280×411. Closing the viewer stopped frames. Wall tile 1.45 s. Agent restart: frames back in 1.3 s, no admin action. Browser share to a selected PC: frames served, Stop → 204. Master-PC share: first frame 0.5 s, 1.9 frames/s, Stop → 204. On-screen notices shown. Audit: 10 views, 9 view ends (one ended by the agent's timeout), 4 shares started / 4 stopped, master set.	**PHYSICAL VALIDATION REQUIRED on a real student PC (IT approval). Admin PC VERIFIED.**
Remote keyboard / mouse	Not implemented, as before. Windows RDP shadowing with the user's consent is documented.	-	**BLOCKED (needs IT authorization)**
Devices offline since 11-12 Oct	Investigated read-only. The router answers, and so does camera 2 (.156). No reply from .150 / .151 / .155 / .102 / .100. The three ESP32 boards went silent in the same minute (12 Oct 23:51): one common cause, unplugged or moved, not a code fault.	Ping and ARP table from this laptop; last-report times from the database.	**PHYSICAL VALIDATION REQUIRED**
Flame detector "17%"	The data shows 642 of its 678 active readings in one block, 9 Oct 16:00-19:00 Cairo time; 11 activations in total; never together with smoke; normal temperature. That fits afternoon sunlight / IR or a bring-up test, not a constantly noisy sensor. No threshold changed. The anomaly finding now states activations and the longest run. This also fixed a real bug: sensor timestamps were read as whole rows.	5 anomaly tests (1 new). Live: "11 separate activations, the longest continuous for 88 min".	**PHYSICAL VALIDATION REQUIRED (bench test in the checklist)**

Incidents (68 open, 63 overdue)	Grouped and investigated; nothing resolved or acknowledged. Mostly bring-up era (2-15 Oct). 20 are two repeated camera warnings from one device. 10 "offline" alerts are correct (those devices are off) and clear by themselves when the devices report again. The page now has "group similar" and shows where each incident came from (software test / unknown origin).	Live grouping query; TypeScript check; deployed.	**PARTIAL: triage needs a person (6 "Forced entry" + 2 "Person with no door entry" first)**
Inventory discrepancy	Unchanged; it waits for a staff count.	The bin still has no baseline count.	**BLOCKED (needs one physical count per bin)**
AI written-answer speed	Profiled: the 1.5B model on the CPU writes about 14 tokens/s, and reading an uncached ~700-token prompt takes about 6 s. Benchmarked prompt sizes on the 13-question set (2 new questions: exact UR5e Home angles, STOP not an E-stop). Adopted 3 passages instead of 4. Added an answer cache: same question + same index replays instantly, and the cache clears on index rebuild.	Facts correct 9/12 → **10/12**. Median first word 8.5 s → **4.4 s**. Median full answer 12.9 s → **7.5 s**; worst case 23.4 s → 12.9 s. Live: STOP question first word 0.95 s, total 3.7 s; repeat instant. Cited lookups 0.1-0.15 s.	**VERIFIED (the model's own wording can still be imprecise; cited lookup stays exact)**
Start-page / analytics speed	Profiled: 10 tiny requests in parallel, served one at a time by a single Python process (12-31 ms each). Cached the backup-status file read (slow Windows bind mount) and throttled the repeated device scan to once per 3 s.	Start page last response 267-509 ms over 3 runs (was ~350-380 ms): **no clear gain**; dominated by CPU contention (cameras, Ollama, test agent). API p95 for all 9 main reads 33-70 ms.	**PARTIAL**
UR5e / FANUC / cameras	No physical access today. UR5e stays in DRY RUN. FANUC stays monitoring-only. Camera 2 live (7 fps); camera 1 has no picture (at the university). Step-by-step lab procedure written: `docs/LAB_READINESS_2026-10-18.md`.	Evidence run: UR5E_01 and FANUC_01 have no joint data; SECCAM_LAB01 degraded.	**PHYSICAL VALIDATION REQUIRED (FANUC interface: BLOCKED)**

2. Tests and evidence

- **Backend:** full suite (result in the reply). New tests: liveness throttle, flame concentration.
- **Tools:** robot monitor 37, knowledge 40, security vision 163, PC agent 5. All passed.
- **Frontend:** TypeScript check clean; production build OK.
- **Live browser tests:** `e2e/pc_screen.mjs` (with the real agent) and `e2e/screen_share.mjs`.
- **Evidence run** `docs/reports/evidence_20261017_135713.md`: 35 pass, 12 fail, 7 not available, 2 simulation. All 12 failures are hardware that's off or away; none is a code defect.

3. Next actions

1. **IT approval:** install the PC agent on one student PC (`tools/pc_agent`: `start_pc_agent.bat` + `agent.env`; Pillow, or OpenCV + numpy). Then test the targeted share to it.
2. **Lab:** power the ESP32 boards and check they come ONLINE; bench-test the flame detector (cover, flame at 50 cm, torch).
3. **UR5e:** cable, Home reached via the pendant, clearance, Remote mode; then your sign-off before `ROBOT_CONTROL=live`.

4. **FANUC:** photograph Version ID / ORDER FI to find the licensed remote option.
5. **Cameras:** check camera 1's picture; mount camera 2; floor calibration; crossing / occlusion tests.
6. **Inventory and incidents:** one count per bin; triage incidents by group with real closure reasons.

4. Files changed in this round

- `tools/pc_agent/pc_agent.py` (version 0.6), `test_pc_agent.py`, `start_pc_agent.bat`, `agent.env.example`, `.gitignore`
- `backend/app/services/anomalies.py`, `devices.py`, `backups.py`; tests `conftest.py`, `test_anomalies.py`, `test_system_health.py`
- `frontend/src/pages/Incidents.tsx`, `components/NetworkedEquipment.tsx`, `lib/api.ts`; `e2e/pc_screen.mjs`
- `tools/knowledge/agent.py`, `serve.py`, `eval_questions.json`
- `docs/LAB_READINESS_2026-10-18.md`, `docs/PROGRESS_CHECKPOINT.md`